Ch. 15 CW #3 – Solutions

Principles of Microeconomics

December 15, 2025

Problems and Applications

9. (a)

Q	MC	VC	TC	ATC
1 pie	\$2	\$2	\$11	\$11
2	4	6	15	7.5
3	6	12	21	7
4	8	20	29	7.25
5	10	30	39	7.8
6	12	42	51	8.5

(b) When price is \$11, the market is in equilibrium at $Q = 200$. Each producer will make 5 pies because P most closely equals MC when $q = 5$ or 6. When $q = 5$, profit is

$$\begin{aligned} Profit &= (P - ATC)Q \\ &= (11 - 7.8)(5) \\ &= 16. \end{aligned}$$

When $q = 6$, profit is

$$\begin{aligned} Profit &= (11 - 8.5)(6) \\ &= 15. \end{aligned}$$

Since they earn more profit when $q = 5$, they'll produce 5.

(c) No, it's not because there are non-zero profits.

(d) In the long run, they make zero profit. Price equals the minimum of average total cost which is approximately \$7 based on part (a). Price equals the minimum of average total cost when firms are producing 3 pies each. Based on the demand schedule, 600 pies are sold in the market which means that there are $600/3 = 200$ producers.

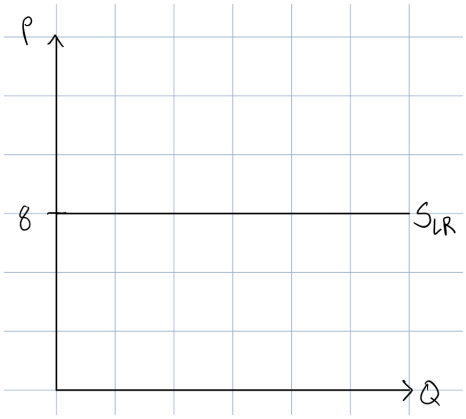
10. (a)

Q	AVC	VC	MC	ATC
1	\$1	\$1	\$1	\$17
2	2	4	3	10
3	3	9	5	8.33
4	4	16	7	8
5	5	25	9	8.2
6	6	36	11	8.67

(b) Each firm produces 5 units because P most closely equals MC when $q = 5$ or 6. Profit is higher when $q = 5$ (\$9 vs. \$8), so the firm produces 5 units. The total quantity supplied is $Q_S = 100q = 500$.

(c) Because profits are positive, firms will enter the market. As the number of suppliers increases, supply will shift right. A rightward shift in supply drives price down and quantity demanded up. The lower price leads to a lower quantity supplied by each firm because price will equal marginal cost at a point further left on the curve.

(d) Based on part (a), the minimum average total cost is approximately \$8.



11. (a) Fixed cost is the portion of total cost that doesn't depend on quantity, \$50. Variable cost is the part of total cost that does depend on quantity $VC = \frac{1}{2}q^2$. Average total cost is

$$\begin{aligned} ATC &= \frac{TC}{q} \\ &= \frac{50 + \frac{1}{2}q^2}{q} \\ &= \frac{50}{q} + \frac{1}{2}q \end{aligned}$$

(b) You can solve this one using calculus or a graph. Alternatively, average total cost is minimized at the point where MC equals average total cost.

$$
\begin{aligned}
MC &= ATC \\
q &= \frac{50}{q} + \frac{1}{2}q \\
\frac{1}{2}q &= \frac{50}{q} \\
\frac{1}{2}q^2 &= 50 \\
q^2 &= 100 \\
q &= 10
\end{aligned}
$$

(c) The firm's supply curve is the marginal cost curve everywhere that $MC \geq ATC$. That means the firm's supply curve is

$$
q = \begin{cases} P, & P \geq 10 \\ 0, & P < 10 \end{cases}
$$

(d) In the short run, firm supply is the marginal cost curve everywhere that $MC \geq AVC$. $AVC = 0.5q$, so $MC \geq AVC$ for all $q \geq 0$. Therefore, market supply is

$$Q_S = 9q = 9P.$$

(e)

$$
\begin{aligned}
Q_S &= Q_D \\
9P_{SR} &= 120 - P_{SR} \\
P_{SR} &= 12 \\
Q_{SR} &= 9(12) \\
&= 108
\end{aligned}
$$

(f) Each firm produces $q_{SR} = P_{SR} = 12$. Each firm's profit is

$$
\begin{aligned}
Profit &= TR - TC \\
&= Pq - (50 + \frac{1}{2}q^2) \\
&= 12(12) - (50 + \frac{1}{2}(12^2) \\
&= 144 - 122 \\
&= 22.
\end{aligned}
$$

Since profit is positive, firms will have an incentive to enter the market.

(g) In the long run, the equilibrium price will be the minimum of ATC, so $P_{LR} = 10$. Market supply will be a horizontal line at $P_{LR} = 10$, so equilibrium quantity will be Q_D when $P = 10$. That's $Q_{LR} = 120 - 10 = 110$.

(h) In the long run, firms still produce the quantity at which $P = MC$, so $10 = q_{LR}$. That means there are $110/10 = 11$ firms in the market.

Non-Mankiw

1. (a)

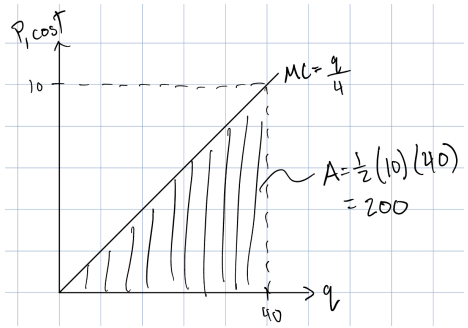
$$Q_D = 10q_D = 10(40 - 2P) = 400 - 20P$$
$$Q_S = 5q_S$$
$$P = \frac{q_S}{4} \implies q_S = 4P \implies Q_S = 5(4P) = 20P$$

$$\begin{aligned} Q_D &= Q_S \\ 400 - 20P^* &= 20P^* \\ 400 &= 40P^* \\ 10 &= P^* \\ Q^* &= 20(10) \\ &= 200 \end{aligned}$$

$$q_S = 4P = 4(10) = 40$$

(b)

$$VC = \sum_0^{40} MC = \text{Area under the MC curve}$$



(c)

$$\begin{aligned} Profit &= TR - TC \\ &= Pq - (FC + VC) \\ &= 40(10) - (50 + 200) \\ &= 150 \end{aligned}$$